



HEALTHCARE SALES & PURCHASE ANALYTICS

Project Report

Data-driven analysis of pharmacy sales, purchasing, products, categories and operational KPIs

Prepared using the supplied healthcare.xlsx dataset

1. Executive Summary

This project analyzes a healthcare/pharmacy transaction dataset containing 42,649 sales records and 1,093 purchase records covering transactions from 02 Jan 2023 to 31 Dec 2024. The analysis focuses on sales performance, purchasing activity, product and category contribution, staff performance, supplier concentration, GST and business KPIs.

Total net sales in the raw sales table are █7,150,609.17, generated from 17,046 bills. Recorded purchases total █3,595,099.30. The strongest category is Respiratory with █1,832,714.50, and the leading product is Salbutamol Inhaler with █959,412.43.

2. Project Objectives

- Measure overall sales and purchasing performance.
- Identify monthly and annual sales trends.
- Determine high-performing healthcare categories and products.
- Compare staff sales contribution.
- Understand supplier purchasing concentration.
- Review sales and purchase GST values.
- Provide actionable recommendations for inventory and business performance.

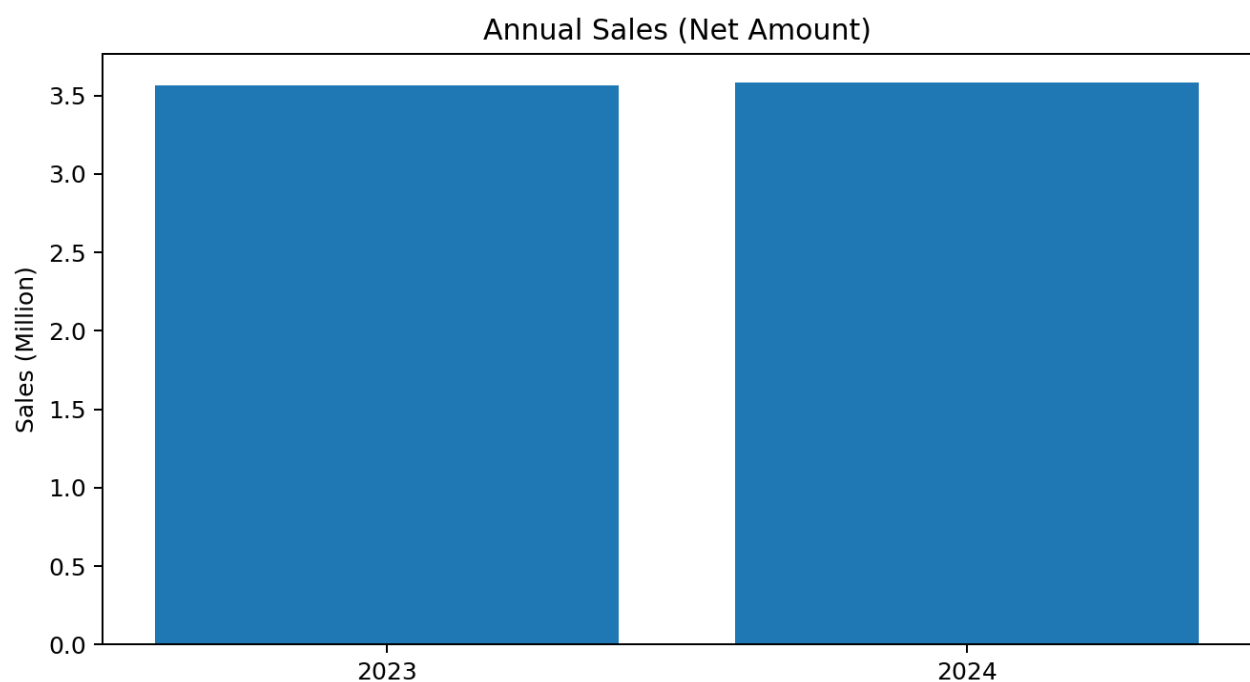
3. Dataset Description

Dataset	Records	Key Fields
weekly_sales	42,649	Bill, Date, Customer, Staff, Product, Category, Qty, GST, Net Amount
weekly_purchase	1,093	GRN, Date, Supplier, Product, Category, Qty, Purchase Price, GST, Total Amount
summary	Pivot/KPI outputs	Monthly, quarterly, category, product and GST summaries
Dashboard	Dashboard area	Workbook dashboard section

4. Key Performance Indicators

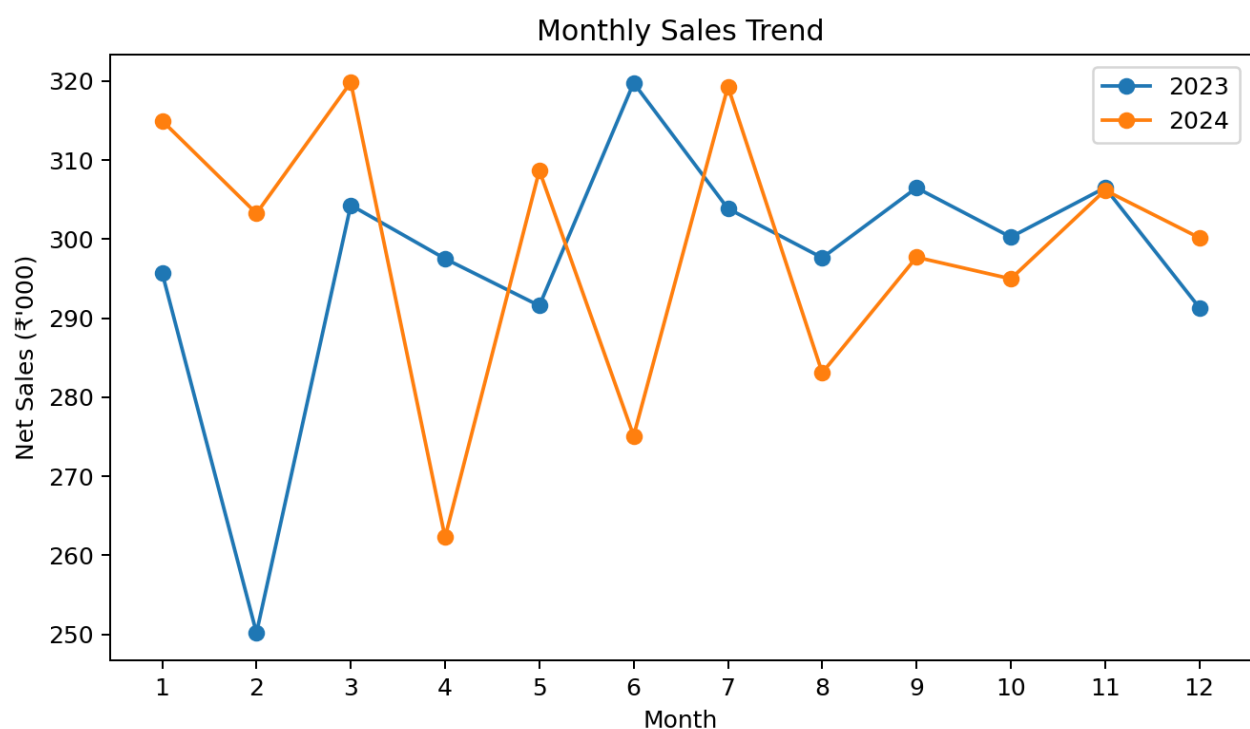
KPI	Value
Total Net Sales	█7,150,609.17
Total Purchases	█3,595,099.30
Sales GST	█748,666.64
Purchase GST	█337,501.74
Sales Quantity	127,775 units
Purchase Quantity	90,459 units
Unique Bills	17,046
Unique Customers	16
Products	25
Categories	13
Staff	4
Suppliers	6

5. Annual Sales Performance



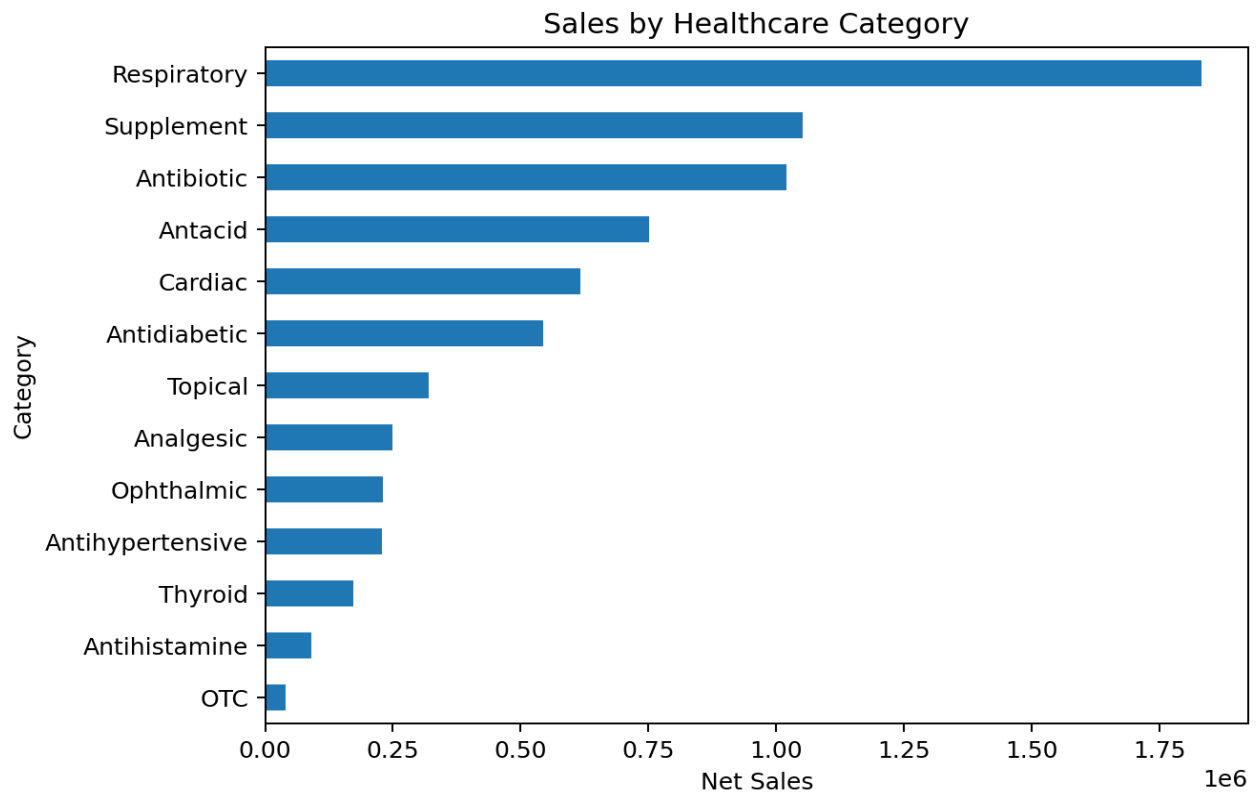
Sales changed from ■3,565,091.20 in 2023 to ■3,585,517.97 in 2024, a year-on-year change of 0.57%. This indicates broadly stable annual sales.

6. Monthly Sales Trend



Monthly analysis helps identify periods of stronger and weaker demand and supports inventory and staffing decisions.

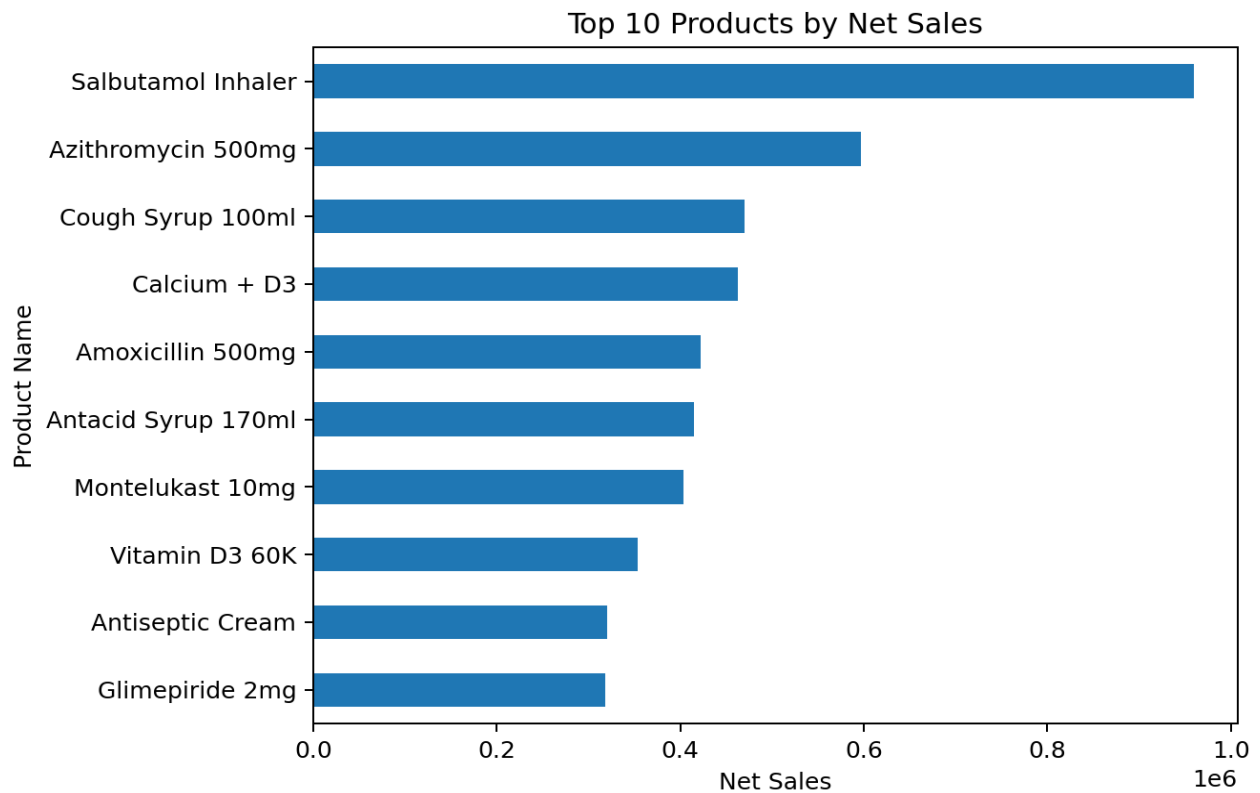
7. Category-wise Sales Analysis



The top three categories—Respiratory, Supplement and Antibiotic—contribute 54.61% of total sales.

Rank	Category	Net Sales	Share
1	Respiratory	■1,832,714.50	25.63%
2	Supplement	■1,052,731.63	14.72%
3	Antibiotic	■1,019,685.64	14.26%
4	Antacid	■751,002.35	10.50%
5	Cardiac	■617,581.33	8.64%
6	Antidiabetic	■544,324.34	7.61%
7	Topical	■320,418.28	4.48%
8	Analgesic	■249,062.39	3.48%

8. Top Product Analysis



Rank	Product	Net Sales	Share
1	Salbutamol Inhaler	■959,412.43	13.42%
2	Azithromycin 500mg	■596,970.11	8.35%
3	Cough Syrup 100ml	■469,921.48	6.57%
4	Calcium + D3	■462,908.67	6.47%
5	Amoxicillin 500mg	■422,715.53	5.91%
6	Antacid Syrup 170ml	■414,895.92	5.80%
7	Montelukast 10mg	■403,380.59	5.64%
8	Vitamin D3 60K	■353,966.23	4.95%
9	Antiseptic Cream	■320,418.28	4.48%
10	Glimepiride 2mg	■318,259.78	4.45%

The top five products account for 40.72% of total sales, so stock availability for these products should be monitored closely.

9. Staff Sales Performance

Rank	Staff	Net Sales	Share
1	Ravi	■1,831,471.20	25.61%
2	Arjun	■1,791,387.17	25.05%
3	Priya	■1,768,218.18	24.73%
4	Meena	■1,759,532.62	24.61%

10. Purchase & Supplier Analysis

The purchase table contains 182 GRNs across 6 suppliers, with total recorded purchases of ₹3,595,099.30.

Rank	Supplier	Purchase Value	Share
1	Sri Balaji Medicals	₹718,800.51	19.99%
2	Sun Pharma Distributors	₹718,248.70	19.98%
3	Karnataka Pharma Dist	₹654,223.73	18.20%
4	Cipla Direct	₹612,818.22	17.05%
5	Medplus Wholesale	₹459,980.38	12.79%
6	Alkem Pharma Dist	₹431,027.76	11.99%

11. GST Analysis

Recorded sales GST is ₹748,666.64 and recorded purchase GST is ₹337,501.74. The difference should not be interpreted as final tax payable without considering input-tax credit rules, adjustments and accounting treatment.

12. Data Reconciliation Observation

The supplied summary sheet contains a separate pre-calculated KPI/pivot section that does not reconcile with the full raw transaction tables. For example, it reports ₹668,154.22 as sales, while the full weekly_sales table totals ₹7,150,609.17. This report therefore uses the complete raw tables for the primary analysis and treats the summary figures as a separate filtered/pivot analysis.

13. Recommendations

- Prioritize inventory planning for Respiratory, Supplement and Antibiotic categories.
- Use reorder thresholds for fast-moving products such as Salbutamol Inhaler and Azithromycin 500mg.
- Monitor batch-level expiry dates, especially for slower-moving products.
- Review supplier concentration and negotiate volume-based pricing.
- Maintain a consistent monthly dashboard for trend monitoring.
- Ensure dashboard KPIs reconcile to the raw source tables.
- Add a clearly defined cost-of-goods measure before making profitability decisions.

14. Limitations

- The dataset is not a complete accounting ledger, so profitability should not be inferred solely from sales and purchase totals.
- The summary sheet contains filtered/pivoted figures that do not reconcile with the full raw tables.
- No external market, competitor or demographic data is included.

15. Conclusion

The Healthcare Sales & Purchase Analytics project converts 42,649 sales and 1,093 purchase records into practical business insights. The analysis identifies stable annual sales, high-value categories, fast-moving products, staff contribution and supplier concentration, providing a strong foundation for inventory planning, sales monitoring and future dashboard development.

16. Suggested Resume Project Title

Healthcare Sales & Purchase Analytics Dashboard

Analyzed healthcare sales and purchase transactions to identify sales trends, top-performing products, category contribution, supplier concentration and GST KPIs using Excel-based data analysis and dashboard reporting.